

CRANE OUTRIGGER SETUP

Ground capacity, mat selection, level and load-chart configuration.

PURPOSE AND SCOPE

Outriggers and stabilizers transfer crane loads to the ground; weak fill, voids, excavations, underground services, slope or inadequate mats can cause overturning. Before setup, authorized persons assess the ground, select the manufacturer-approved configuration, support all beams and jacks, and keep the crane within load-chart level limits.

KEY HAZARDS

 OVERTURN	 WEAK GROUND	 RESTRICTED SETUP	 CRUSH ZONE
 WRONG CONFIGURATION	 RAIN / SETTLEMENT	 INADEQUATE MAT	 UNDERGROUND VOID

CRITICAL VALUES TO VERIFY IN THE FIELD

100%

Stabilizer configuration matching the load chart

0 gaps

Full contact between mat and jack

Continuous

Monitoring of level and ground settlement



REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA Subpart CC links ground conditions, manufacturer procedures, setup control and shift inspection. Ground bearing capacity must be verified with project data.

OSHA 1926.1402

GROUND CONDITIONS

Ground must have adequate firmness, drainage and grading to support the equipment; known underground hazards must be communicated.

OSHA 1926.1404

MANUFACTURER PROCEDURES

Assembly and disassembly must follow manufacturer procedures or employer procedures meeting qualified-person requirements.

OSHA 1926.1412(d)

SHIFT INSPECTION

A competent person must evaluate ground, level, stabilizers and visible equipment condition before each shift.

YÜK ÇİZELGESİ / ÜRETİCİ

CONFIGURATION MATCH

Outrigger extension, counterweight, boom and working radius must exactly match the load chart in use.

MINIMUM EVIDENCE BEFORE START

- | | |
|--|--|
| ☐ Ground-bearing verification | ☐ Underground-services plan |
| ☐ Mat / cribbing calculation | ☐ Load-chart configuration |
| ☐ Shift inspection record | ☐ Lifting plan |

SUPERVISOR DISCUSSION POINTS

- What reaction load was used to select this mat?
- Who reassesses the ground after rain?
- How will outrigger settlement be detected during the lift?

OUTRIGGER, MAT AND GROUND INSPECTION



Firm ground - Correct mat - Full contact - Locked beam - Level - Barricade

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. RELOCATE THE SETUP

Choose a wider, firmer area away from excavations, voids and weak fill.

2. ENGINEER THE GROUND SUPPORT

Verify bearing capacity, underground services, mat size and load distribution.

3. SET THE CORRECT CONFIGURATION

Extend, lock and fully support every beam and jack to manufacturer procedures.

4. MONITOR LEVEL AND SETTLEMENT

Monitor crane level, mat movement, water and ground cracking throughout the lift.

CRITICAL WARNING



Never set an outrigger jack on bricks, loose timber, covers or unknown fill. If a mat settles, bends or slips, or the ground cracks, stop the lift and place the load in a safe position.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|--|---|
| ☐ Ground assessment complete | ☐ Underground services and voids known |
| ☐ Mat size and material suitable | ☐ All outriggers match the load chart |
| ☐ Beams extended and locked | ☐ Jacks fully contact mats |
| ☐ Crane level within manufacturer tolerance | ☐ Crush zone barricaded |
| ☐ Settlement and weather monitoring planned | |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			